Personal Project Report

Smart Pumping Station

PLC Control, HMI, Reliability Monitoring and Explainable Anomaly Detection

with TIA Portal, S7-1200 and OPC UA

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This report describes a personal project conducted independently.

Repository: https://github.com/HoraEmbedded/smart-pumping-station

Résumé

Ce rapport présente la conception, la réalisation et la vérification en simulation d'une station de pompage d'utilité industrielle. Deux pompes, l'une de service et l'autre de secours, maintiennent un réservoir entre deux niveaux d'eau. Un automate Siemens S7-1200 programmé sous TIA Portal V18 gère les modes de fonctionnement, les interverrouillages, l'alternance des pompes, les diagnostics et les alarmes. Une IHM WinCC Runtime Professional offre cinq écrans et un système d'alarmes natif, avec un bouton d'arrêt d'urgence dont le relâchement est confirmé. Un collecteur OPC UA en lecture seule alimente des indicateurs de fiabilité, un tableau de bord local et six règles explicables de détection d'anomalies. La démarche suit un cycle en V itératif avec 56 exigences, 69 tests, une matrice de traçabilité dans les deux sens et un journal de décisions de 61 entrées. L'ensemble est validé en simulation.

Mots-clés : automatisation industrielle, API, TIA Portal, S7-1200, OPC UA, IHM, fiabilité, détection d'anomalies.

Abstract

This report describes the design, implementation and verification in simulation of an industrial utility pumping station. Two pumps, one duty and one standby, keep a storage tank between two water levels. A Siemens S7-1200 controller programmed in TIA Portal V18 handles operating modes, interlocks, pump alternation, diagnostics and alarms. A WinCC Runtime Professional HMI gives the operator five screens and a native alarm system, with an emergency stop button whose release requires confirmation. A read-only OPC UA collector records datasets that feed reliability indicators, a local dashboard and six explainable anomaly rules. The work follows an iterative V-model with 56 requirements, 69 tests, a bidirectional traceability matrix and a decision log of 61 entries. It is validated in simulation.

Keywords: industrial automation, PLC, TIA Portal, S7-1200, OPC UA, HMI, reliability, anomaly detection.

Contents

1 Industrial context and need **1**

2 Requirements and method **2**
2.1 Iterative V-model . . . 2
2.2 Requirements . . . 2
2.3 Traceability in both directions . . . 2

3 Modeling assumptions **3**
3.1 Volume balance . . . 3

4 System modelling **5**
4.1 Context diagram . . . 5
4.2 Use cases . . . 5
4.3 Functional decomposition . . . 6
4.4 Sequence diagram: pump failure (UC-03) . . . 6
4.5 Link with the following chapters . . . 7

5 Process architecture and P&ID **8**

6 I/O list and alarms **9**
6.1 I/O list . . . 9
6.2 Alarm philosophy . . . 9

7 Operating modes and Grafcet **10**
7.1 Interlocks . . . 11
7.2 Faults, reset and pumps . . . 12

8 PLC architecture **13**

9 HMI and supervision **14**

10 Data, indicators and anomaly detection **16**
10.1 Data architecture . . . 16
10.2 Reliability indicators and dashboard . . . 16
10.3 Explainable anomaly rules . . . 18

11 Test plan and results **21**
11.1 Method . . . 21
11.2 Acceptance scenarios . . . 21
11.3 Traceability and defects . . . 22
11.4 Reproducibility . . . 22

12 Limitations and perspectives **23**

General conclusion 24

References 25

A Repository structure 26

B Main parameters 27

C Alarm list 28

List of Figures

3.1 Reference cycle of the tank level between the start and stop thresholds. 4

4.1 Context diagram of the smart pumping station. Commands flow from the actors to the controller; measurements flow from the process; datasets and indicators cross the OT/IT boundary in read-only mode. 5
4.2 Sequence diagram of UC-03: pump 1 fails while running, pump 2 takes over within a second, and the alarm reaches the HMI. 7

5.1 Simplified P&ID with the PLC tag name of every signal. 8

7.1 Grafcet of the station: modes (G1) and supervisor (G0) forcing the stop and the fault. 11
7.2 Grafcet of the automatic pumping cycle, enclosed in the automatic state. 12

9.1 Overview screen with the station running. 14
9.2 Alarm screen with an active alarm. 15

10.1 OT and IT architecture: the collector only reads. 16
10.2 Local read-only dashboard in live mode. 18
10.3 Signals and flagged periods for the leak dataset. 20

List of Tables

1.1 Iterations and releases. . . . 1

2.1 The 56 requirements by group. . . . 2

3.1 Simulation assumptions. . . . 3
3.2 Reference scenario computed from the volume balance. . . . 3

4.1 The five use cases and their acceptance tests. . . . 6
4.2 Functional layers of the station. . . . 6

5.1 Instruments and their PLC tags. . . . 8

6.1 The 13 alarms by class. The full list is in Appendix C. . . . 9

7.1 Station states. . . . 10
7.2 Interlocks. . . . 11

8.1 The 14 code blocks. . . . 13

9.1 HMI screens. . . . 14

10.1 The six datasets. Ground-truth columns are kept apart from the measurements. . . . 16
10.2 Indicator definitions. . . . 17
10.3 Indicators of the six datasets (from `kpi_summary.md` and `kpi_summary.csv`). . . . 17
10.4 Anomaly rules. . . . 18
10.5 Rule outcome per dataset (TP: expected rule fired, ok: neither expected nor fired). . . . 19
10.6 Detection against the injected faults (from `anomaly_eval.csv`). . . . 19
10.7 Threshold margins on the normal dataset. . . . 19

11.1 Results of the final campaign (from `campaign_summary.md`). . . . 21
11.2 End-to-end acceptance tests. . . . 21
11.3 Examples from the traceability matrix. . . . 22
11.4 Defects found by execution. . . . 22

List of abbreviations

API	Automate Programmable Industriel
DB	Data Block
DI, DQ	Digital Input, Digital Output
FB	Function Block
FC	Function
GRAFCET	Graphe Fonctionnel de Commande des Étapes et Transitions
HMI	Human-Machine Interface
I/O	Inputs and Outputs
IEC	International Electrotechnical Commission
ISA	International Society of Automation
ISO	International Organization for Standardization
KPI	Key Performance Indicator
LAD	Ladder Diagram
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
OB	Organization Block
OPC UA	Open Platform Communications Unified Architecture
PLC	Programmable Logic Controller
P&ID	Piping and Instrumentation Diagram
SCADA	Supervisory Control and Data Acquisition
SCL	Structured Control Language
SysML	Systems Modeling Language
TIA	Totally Integrated Automation
UML	Unified Modeling Language

CHAPTER 1

Industrial context and need

A factory utility station moves water from a source to a storage tank that feeds the plant. The station must keep the tank between a low and a high level while avoiding four situations: a pump running dry when the source is empty, too many starts that wear motors and contactors, an overflow of the tank, and an unsafe restart after a critical fault.

This project treats the station as a complete engineering case, from the requirements to the verification. It covers:

- the process study, the P&ID, the I/O list and the alarm philosophy;
- requirements, a test plan written before the code and a traceability matrix;
- the PLC logic: state machine, level control, two-pump alternation, interlocks and diagnostics;
- the operator HMI with a native alarm system and an emergency stop whose release is confirmed;
- data export, reliability indicators, a local dashboard and explainable anomaly rules;
- a final test campaign on a frozen build and five end-to-end acceptance scenarios.

The project is delivered in fifteen iterations, each closed by a tagged release (Table 1.1).

Iteration	Tag	Content
1	v0.1.0	Repository foundation, scope, simulation assumptions
2	v0.2.0	Process study, P&ID, I/O list, alarm matrix
3	v0.3.0	Requirements, traceability matrix, test plan before coding
4	v0.4.0	Grafcet and state machine
5	v0.5.0	TIA Portal project and level control
6	v0.6.0	Modes, interlocks, latched critical faults
7	v0.7.0	Two pumps, alternation, backup
8	v0.8.0	Diagnostics and monitoring
9	v0.9.0	HMI overview and commands
10	v0.10.0	Alarm system, maintenance and trend screens, operator guide
11	v0.11.0	Data export and datasets
12	v0.12.0	Indicators and dashboard
13	v0.13.0	Explainable anomaly rules
14	v0.14.0	Final test campaign and regression
15	v1.0.0	Report, demonstration video, packaging, HMI emergency stop

Table 1.1. Iterations and releases.

CHAPTER 2

Requirements and method

2.1 Iterative V-model

Each version is a small V cycle: requirements, design, implementation, then tests that mirror each design level. The final project has 56 requirements and 69 tests. Every requirement is linked to its design reference and its tests in a traceability matrix, checked by script in both directions: no requirement without a test, no test absent from the matrix. Every decision is recorded in a log of 61 entries with its context, the alternative rejected and the consequence.

2.2 Requirements

Requirements are written with “shall”, one idea each, with a measurable condition and a verification method: test, inspection, analysis or demonstration. Two examples:

- **REQ-011:** The simulated emergency stop shall stop all pump commands and latch the station in FAULT.
- **REQ-056:** The HMI shall provide an emergency stop request that stops all pumps on a single action without confirmation, acts as the emergency stop contact, and stays engaged until released after a confirmation.

Group	Requirement identifiers
Control and modes	REQ-001 to 009, 033 to 036
Interlocks and faults	REQ-010 to 015
Diagnostics and counters	REQ-016 to 020, 037 to 040
HMI and alarms	REQ-021 to 023, 056
Data and analytics	REQ-024 to 027, 049 to 054
Quality and release	REQ-028 to 032, 055

Table 2.1. The 56 requirements by group.

The first version had 32 requirements. The others came out of the design and implementation work (state machine, pump logic, HMI interface, diagnostics, emergency stop request), each added with an identifier, a test and a matrix line.

2.3 Traceability in both directions

The traceability matrix links every requirement to its design reference and to its tests. A script checks both directions: no requirement without a test, no test absent from the matrix. When the code changes, an impact analysis script lists the tests to replay from the matrix, and a shared block such as the HMI interface data block triggers a full level 1 regression.

CHAPTER 3

Modeling assumptions

The process is simulated inside the PLC (an interrupt of 100 ms), so that faults can be injected reproducibly and the same program can later read real inputs. The model rests on the assumptions of Table 3.1.

Item	Assumption
Storage tank	Volume 10 m^3, level normalized from 0 to 100 %
Pump	Delivers 12 m^3/h while its run feedback is on and the source is available
Consumption	Constant drain of 3 m^3/h
Motor current	Nominal 10 A, multiplied by 1.3 when a high-current fault is injected
Electrical power	$P = \sqrt{3}\,U\,I\cos\varphi/1000$ in kW, $U = 400$ V, $\cos\varphi = 0.85$
Analog signals	4–20 mA, ranges realistic but not calibrated for a physical plant
Time	Process dynamics can be accelerated (factor 60), protection delays run in PLC real time

Table 3.1. Simulation assumptions.

3.1 Volume balance

The tank level follows a volume balance. With flows in m^3/h, one explicit Euler step of $\Delta t = 0.1$ s gives the update executed by the PLC:

$$\frac{dV}{dt} = Q_{in} - Q_{out}, \qquad L_{k+1} = L_k + \Delta t\,\frac{100\,(Q_{in} - Q_{out})}{3600\,V_{tank}} \tag{3.1}$$

Quantity	Value
Rise rate, one pump running	0.025 %/s (1.5 %/min)
Fall rate, no pump	0.0083 %/s (0.5 %/min)
Fill time, 30 to 80 %	about 33 min
Drain time, 80 to 30 %	about 100 min
Cycle duration	about 133 min
Pump duty	about 25 %
Reference start rate	about 0.45 starts per hour

Table 3.2. Reference scenario computed from the volume balance.

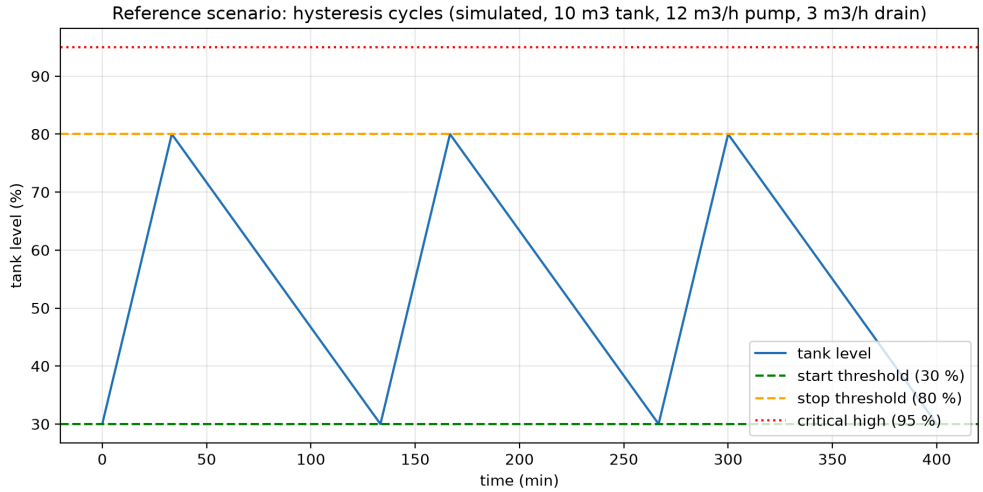


Figure 3.1. Reference cycle of the tank level between the start and stop thresholds.

The cycle time depends on the consumption. With a band volume V_{band} between the two thresholds,

$$T_{cycle} = \frac{V_{band}\, Q_{pump}}{Q_{out}\,(Q_{pump} - Q_{out})}, \qquad T_{min} = \frac{4\,V_{band}}{Q_{pump}} \text{ at } Q_{out} = \frac{Q_{pump}}{2} \tag{3.2}$$

For $V_{band} = 5$ m^3 and $Q_{pump} = 12$ m^3/h the shortest healthy cycle lasts 100 minutes, so a healthy station never starts more than 0.6 times per hour. The alarm threshold of 3 starts per hour is five times this maximum, which makes it a real anomaly and not a consumption fluctuation.

CHAPTER 4

System modelling

This chapter presents the station as a system: the actors around it, the five use cases that structure the design, the functional decomposition between the PLC, the HMI, the collector and the rules, and one sequence diagram for the most critical case.

4.1 Context diagram

The station sits between the process (the water it moves) and the operators (the people who use and maintain it), and it exchanges data with the plant information layer. Figure 4.1 shows the actors and the main flows. The dashed line is the OT/IT boundary: to the left, the process and the controller; to the right, the monitoring and analytics that read but never write.

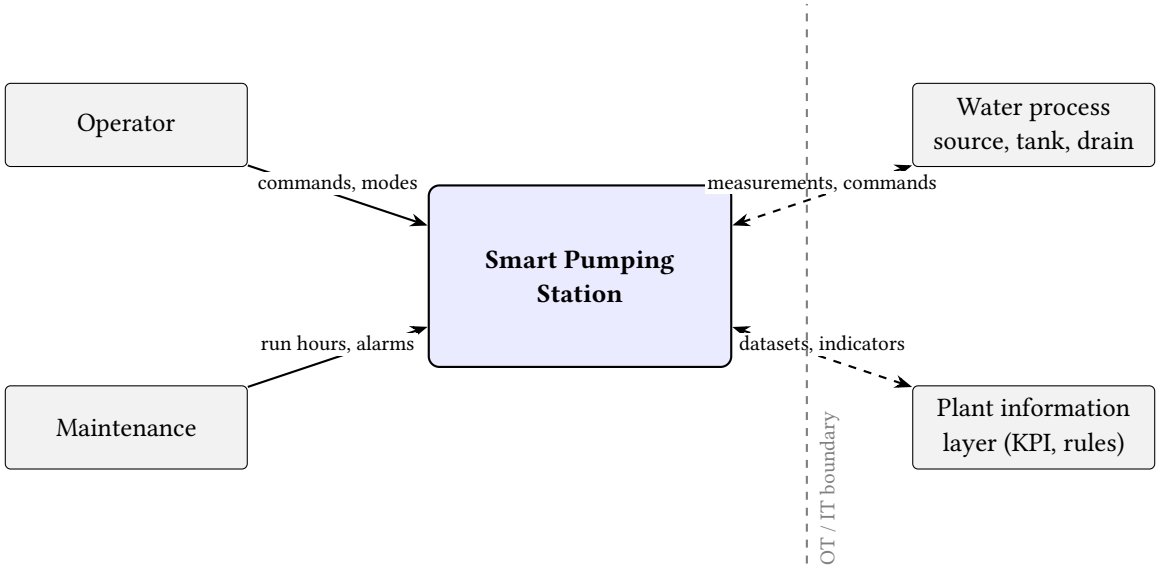


Figure 4.1. Context diagram of the smart pumping station. Commands flow from the actors to the controller; measurements flow from the process; datasets and indicators cross the OT/IT boundary in read-only mode.

4.2 Use cases

The design is organised around five use cases, each with a nominal scenario and an acceptance test that replays it end to end (Chapter 11). Table 4.1 summarises them.

ID	Actor	Nominal scenario	Test
UC-01	Operator	Start the station in Auto, the priority pump starts at the low level and the level rises	T61
UC-02	Operator	Continue until the high level, the pump stops and the priority passes to the other pump	T62
UC-03	Process	A pump fails during operation, the backup takes over and the station keeps running	T63
UC-04	Process	The source runs low, the pump stops, starts are inhibited until the source recovers	T64
UC-05	Process	A blocked flow is detected after the validation delay, the pump is locked out, the backup takes over	T65

Table 4.1. The five use cases and their acceptance tests.

4.3 Functional decomposition

The five use cases are realised by four functional layers, each with its own responsibility (Table 4.2). This separation is what allows the anomaly rules to be added later without touching the PLC program.

Layer	Responsibility	Implementation
Control	Level control, pump choice, alternation, interlocks, state machine, latched faults	PLC, SCL and Ladder
Supervision	Screens, commands, alarms, emergency stop request with confirmed release	WinCC Runtime
Acquisition	Read the export block once per second, write a CSV with the PC timestamp	OPC UA collector, Python
Analysis	Indicators, dashboard, explainable anomaly rules on the recorded data	Python scripts, offline

Table 4.2. Functional layers of the station.

4.4 Sequence diagram: pump failure (UC-03)

Figure 4.2 shows the most critical use case: a pump fails while it is the one running. The operator does not act; the process and the PLC handle the situation. The station must keep running, the backup pump must take over within a second, and the alarm must appear in the HMI.

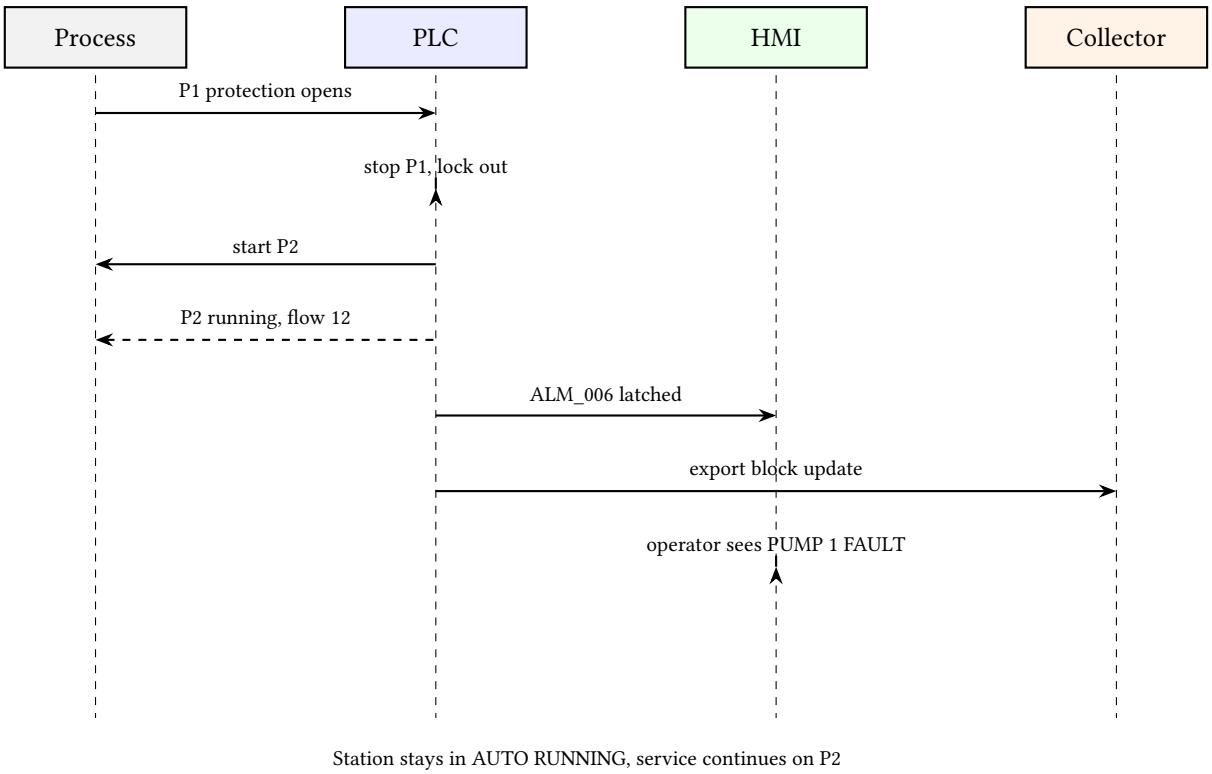


Figure 4.2. Sequence diagram of UC-03: pump 1 fails while running, pump 2 takes over within a second, and the alarm reaches the HMI.

4.5 Link with the following chapters

The context, the use cases and the layers above drive the chapters that follow. Chapter 5 describes the process architecture; Chapter 6 lists the I/O and the alarms; Chapter 7 presents the operating modes and the Grafcet; Chapter 8 describes the PLC blocks; Chapter 9 presents the HMI; Chapter 10 covers the data path, the indicators and the rules; Chapter 11 reports the final campaign.

CHAPTER 5

Process architecture and P&ID

Water flows from a low source tank through one of two parallel pumps to a storage tank, then to a simulated consumption. Each pump has suction and discharge isolation valves, a check valve prevents backflow, and the instruments follow the ISA 5.1 tag convention.

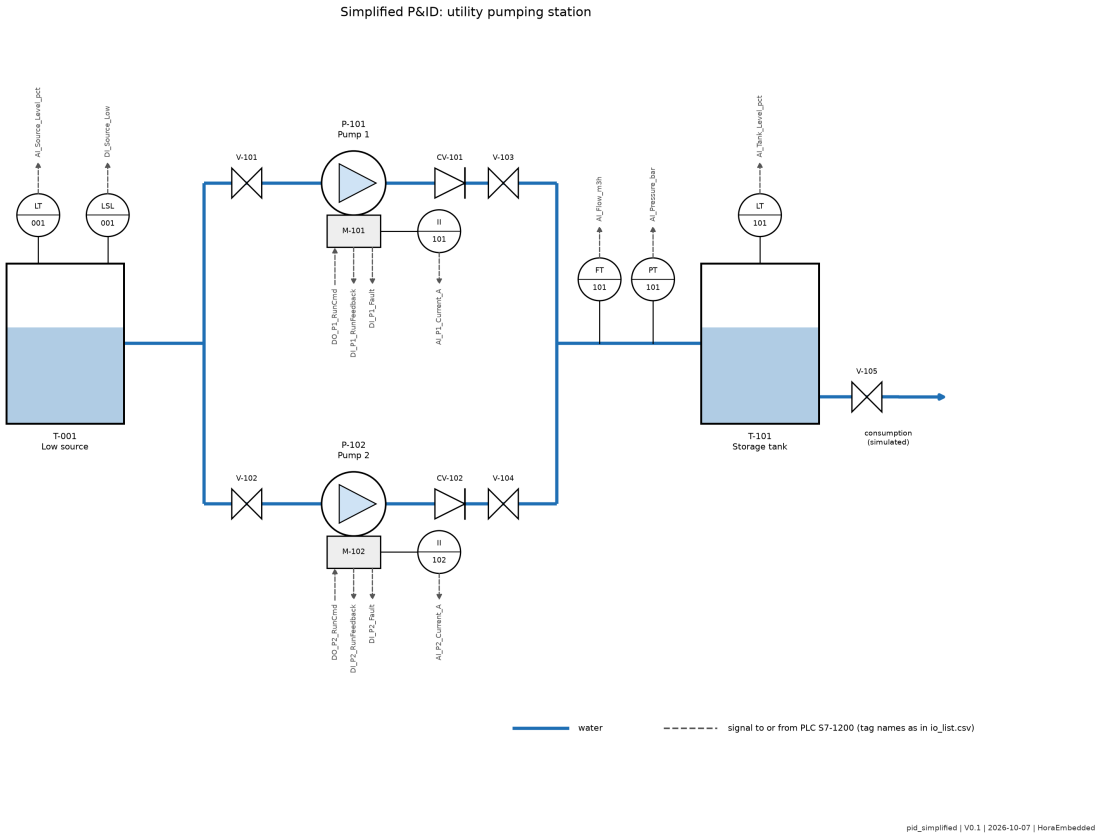


Figure 5.1. Simplified P&ID with the PLC tag name of every signal.

Tag	Measurement	PLC tag
LSL-001, LT-001	Source level, switch and analog	`DI_Source_Low`, `AI_Source_Level_pct`
LT-101	Tank level	`AI_Tank_Level_pct`
FT-101	Discharge flow	`AI_Flow_m3h`
PT-101	Discharge pressure	`AI_Pressure_bar`
II-101, II-102	Motor currents	`AI_P1_Current_A`, `AI_P2_Current_A`

Table 5.1. Instruments and their PLC tags.

CHAPTER 6

I/O list and alarms

6.1 I/O list

The I/O list is the contract between the process, the PLC, the HMI and the tests: 24 signals, 12 digital inputs, 6 analog inputs and 6 digital outputs. Nothing is programmed without being listed there, and no raw address appears in the code.

Analog scaling. A 4–20 mA signal is converted to an integer from 0 to 27648:

$$x = x_{min} + \frac{raw}{27648}\,(x_{max} - x_{min}), \qquad mA = 4 + \frac{raw}{1728} \tag{6.1}$$

The live zero at 4 mA separates a measurement at zero from a cut wire: a raw value under the low limit (default -500 counts, about 3.7 mA) marks the signal invalid.

Fail-safe wiring. The emergency stop and the pump protection contacts are normally closed, so a cut wire and a pressed button give the same result, the stop.

6.2 Alarm philosophy

The alarm design follows the principles of ISA-18.2 at the scale of this station:

- every alarm requires an operator action, otherwise it is a message;
- every alarm has a priority, probable causes, a system reaction, an HMI text and a reset condition;
- critical faults latch, and nothing restarts automatically after one;
- a dependent alarm is suppressed when a higher one already explains it.

Class	Alarms	Acknowledgment
Critical	4	Required
High	3	Required
Medium	2	Required
Warning	4	Not required, clears by itself

Table 6.1. The 13 alarms by class. The full list is in Appendix C.

CHAPTER 7

Operating modes and Grafcet

The station is always in exactly one of seven states (Table 7.1).

Code	State	Pumps
0	STOPPED	Off, known state
10	INITIALIZING	Off, inputs checked for 2 s
20	READY	Off, waiting for a mode
30	AUTO_RUNNING	Automatic level control
40	MANUAL_RUNNING	On operator command
50	MAINTENANCE	Pump tests limited to 60 s
90	FAULT	Off, waiting for correction and reset

Table 7.1. Station states.

The numbers step by ten so that a state can be inserted without renumbering the HMI and the exports. Every mode change passes through READY, so the pumps stop before the new mode starts. If two mode positions are active at once, the station counts it as no selection and stays in READY.

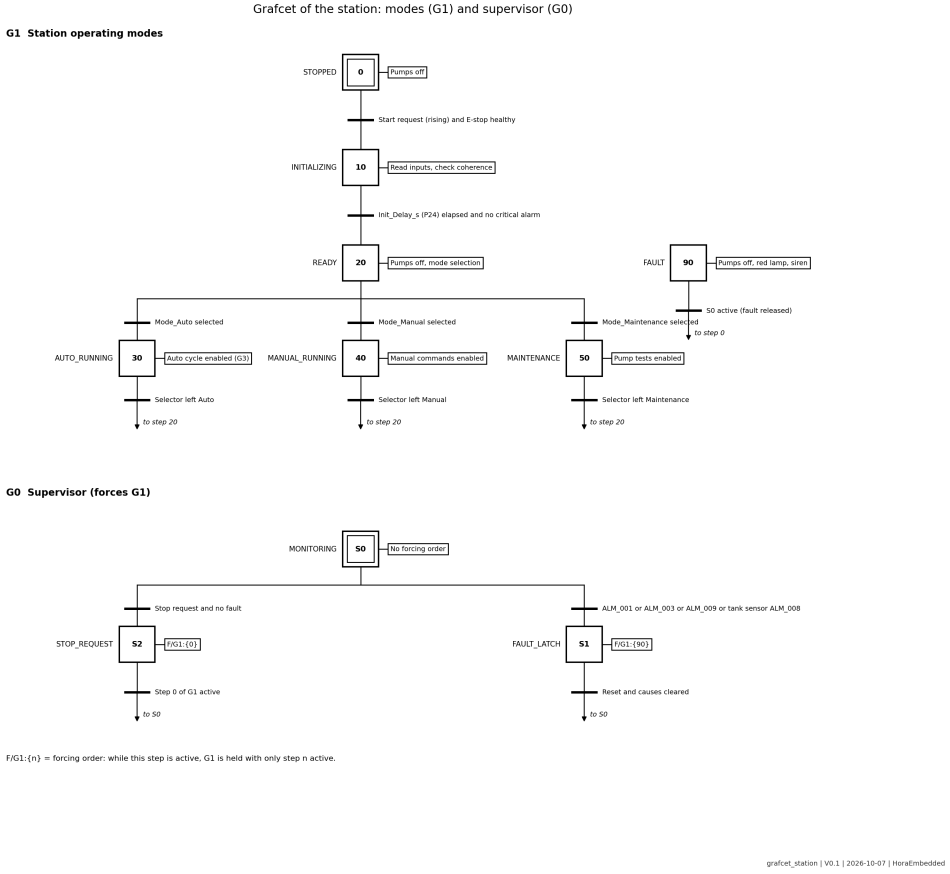


Figure 7.1. Grafcet of the station: modes (G1) and supervisor (G0) forcing the stop and the fault.

The Grafcet (IEC 60848) uses a supervisor G0 that forces G1 to step 0 on a stop request and to step 90 on a critical fault, so that stop and fault always win without an arrow from every step.

7.1 Interlocks

Interlocks act in every state and every mode. Manual and maintenance requests go through the same path to the pump output as the automatic request, so none of them bypasses an interlock.

Interlock	Effect	Alarm
Emergency stop contact open, or HMI request engaged	All pumps off, FAULT latched	ALM_001
Source low, with hysteresis	Pumps stop, starts inhibited	ALM_002
Tank level above the critical threshold	All pumps off, FAULT latched	ALM_003
Pump protection open or no run feedback	That pump unavailable	ALM_006, 007
Low flow after the validation delay	That pump unavailable	ALM_004, 005
Demand and no pump available	FAULT latched	ALM_009
Tank level signal invalid	FAULT latched	ALM_008
Another pump already running	Second run command refused	none
Restart delay not elapsed	Start refused	none

Table 7.2. Interlocks.

7.2 Faults, reset and pumps

A fault is stored in two variables: the cause, which follows the process, and the latch, which remembers it. The reset acts on a rising edge and only clears a latch whose cause is gone. After a valid reset the station returns to STOPPED and a new start request is required, so nothing restarts by itself.

Only one pump runs at a time. The priority pump changes after each completed normal cycle, start at the low threshold and stop at the high threshold without an interruption by a fault. If the priority pump is unavailable the other one is used, also in the middle of a cycle, and an interrupted cycle does not swap the priority.

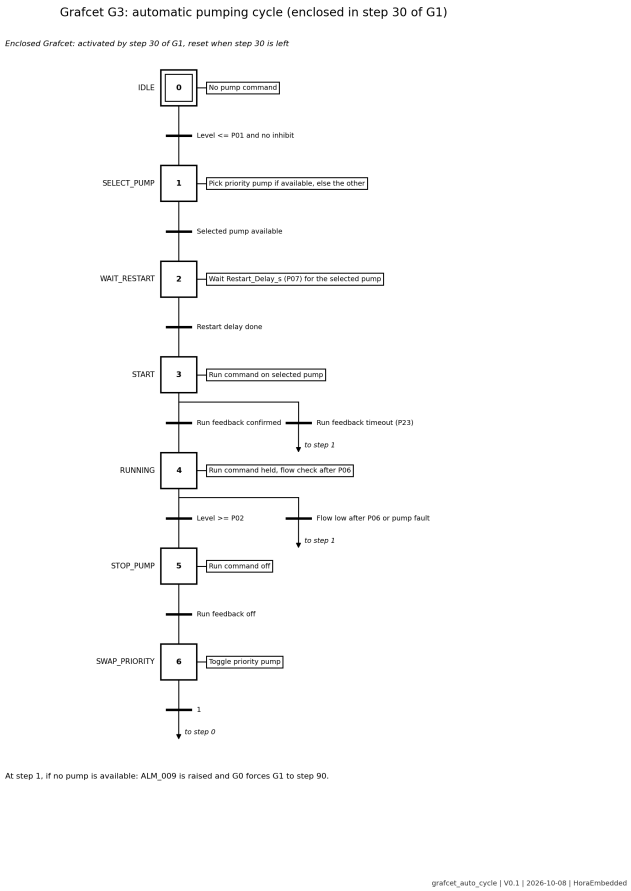


Figure 7.2. Grafcet of the automatic pumping cycle, enclosed in the automatic state.

CHAPTER 8

PLC architecture

The controller is a Siemens S7-1200, CPU 1214C DC/DC/DC with two 4-channel analog input modules, programmed in TIA Portal V18 and simulated with S7-PLCSIM V18 SP2. Languages are chosen block by block: Ladder for the latched alarms and the indicators, which maintenance staff read, and SCL for the state machine and the calculations, because it exports as plain text and gives readable differences in Git.

Block	Language	Role
`OB1`	SCL	Orchestration only
`OB30`	SCL	Cyclic interrupt, 100 ms, process model
`FB_ModeManager`	SCL	State machine
`FB_AutoCycle`	SCL	Level demand, pump choice, alternation, backup
`FB_PumpControl`	SCL	Run command, restart delay, flow and feedback checks, counters
`FB_StationMonitor`	SCL	Frequent starts, maintenance due, level rise, sensor warnings
`FC_ScaleAnalog`	SCL	4–20 mA scaling and validity
`FC_ReadAndScaleInputs`	SCL	Effective process values, from the model or the inputs
`FC_Interlocks`	SCL	Fault causes, source availability, HMI E-stop request
`FC_AlarmLatch`	Ladder	Latched alarms and conditional reset
`FC_Indicators`	Ladder	Lamps and siren
`FC_Interfaces`	SCL	HMI interface, export block, operator messages
`FC_WriteOutputs`	SCL	Pump outputs
`FC_SimulationModel`	SCL	Tank model, fault injection

Table 8.1. The 14 code blocks.

Seven global data blocks hold the parameters, the effective process values, the simulation, the station state, the alarms, the HMI interface and the export block. `OB1` calls the blocks in this order: input scaling, interlocks, alarm latches, mode manager, automatic cycle, pump 1, pump 2, station monitor, interfaces, outputs, indicators. Pump 2 is evaluated after pump 1 and reads its command, so when both are requested in the same scan, pump 1 starts and pump 2 is refused.

All thresholds and delays live in the parameter data block (Appendix B). The SCL sources are the reference and are versioned in text. The binary TIA project is archived in each GitHub release.

CHAPTER 9

HMI and supervision

The operator interface is built in WinCC Runtime Professional V18 and simulated against the PLC simulation. It reads and writes only the HMI interface data block, filled by one function, so the screens do not depend on the internal block structure.

Screen	Purpose
Overview	Source, pumps with command and feedback shown apart, tank, measurements, global state
Commands	Start, stop, reset, mode selection, manual pump commands, silence, emergency stop
Alarms	Active alarms with class, time and info text
Maintenance	Run hours, start counts, current, service status, validity of every signal
Trends	Tank level, flow, pressure and active pump over time

Table 9.1. HMI screens.

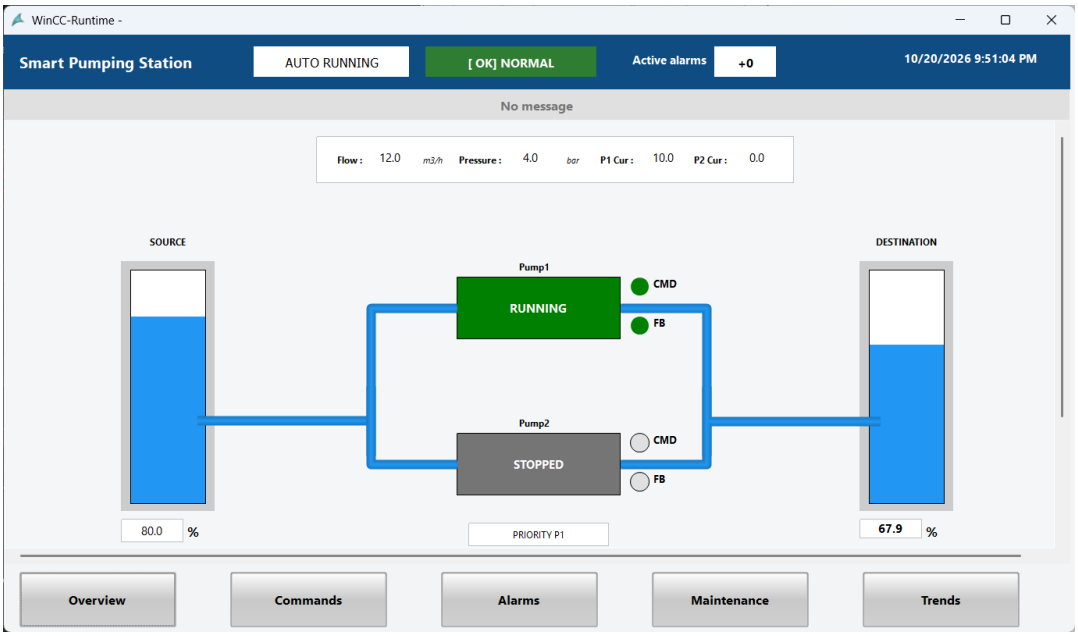


Figure 9.1. Overview screen with the station running.

Design rules: every state is written as text with a symbol, so the colour is a second channel, and a pump command without feedback shows STARTING, then UNAVAILABLE after the timeout.

The native WinCC alarm system carries 13 discrete alarms in four classes, triggered by the bits of two status words. Each alarm has an info text with probable causes and the action to take. Acknowledging means “seen” and never clears a fault: only a reset with the cause removed does, and the screen says why when it refuses. The siren can be silenced and sounds again for any new critical alarm.

An emergency stop request is available on every screen: it acts on a single press without confirmation, like the emergency stop contact, and its release asks for a confirmation. The emergency stop button is red, large and apart from the other commands, so that a single press stops all pumps at once, latches ALM_001 and puts the station in FAULT.

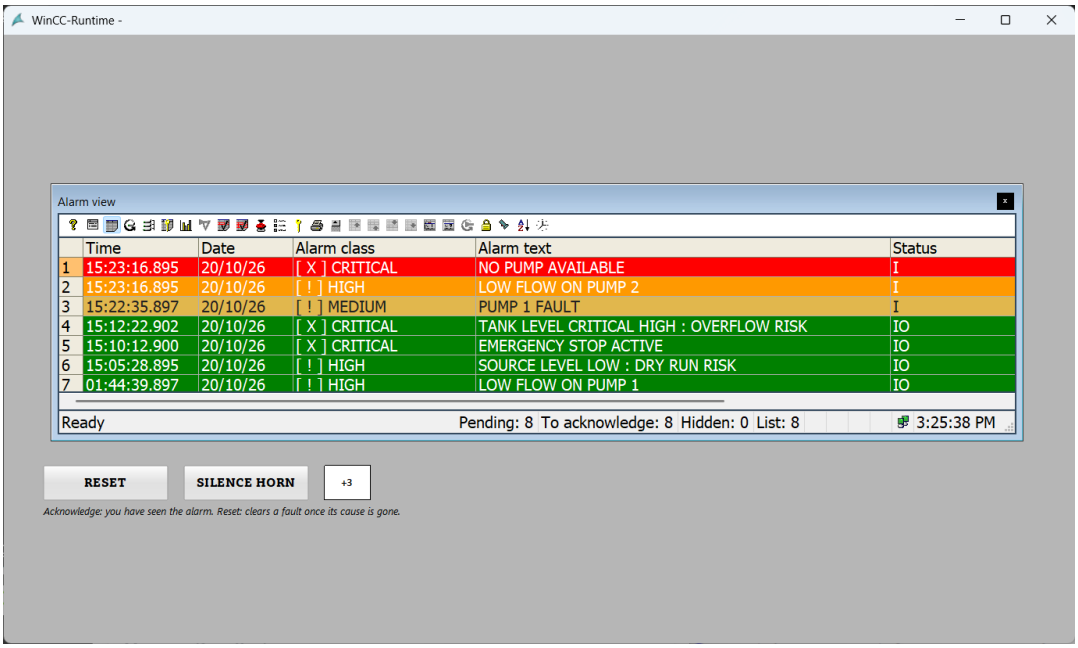


Figure 9.2. Alarm screen with an active alarm.

CHAPTER 10

Data, indicators and anomaly detection

10.1 Data architecture

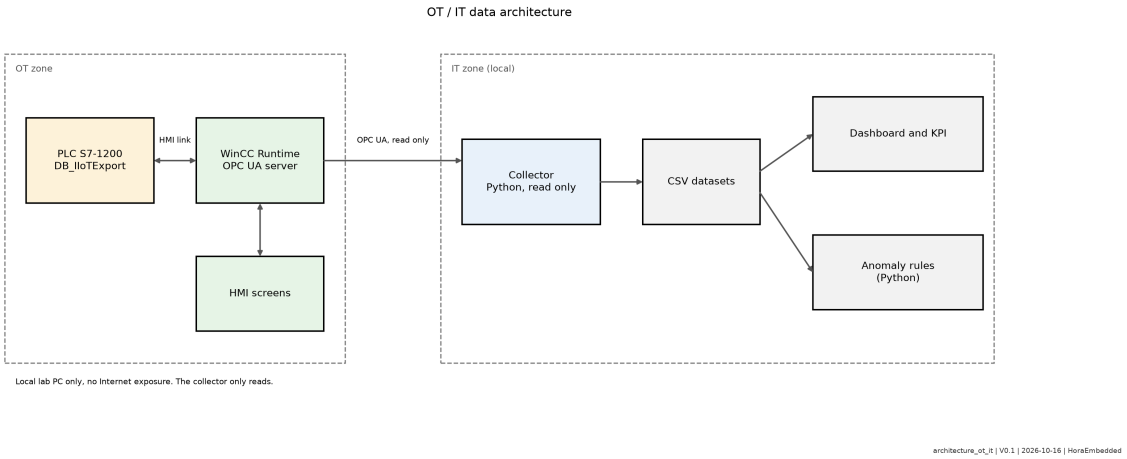


Figure 10.1. OT and IT architecture: the collector only reads.

The PLC publishes a flat export block (70 bytes, fixed layout). It carries a sample counter that increases once per real second and the simulated clock. The two clocks matter: protection delays run in real PLC seconds, while process dynamics and rates run on simulated time.

Route taken. The first plan was to read the PLC directly from Python. A short feasibility test showed that the advanced simulator does not support the S7-1200 and that the standard one cannot be reached from outside. After comparing alternatives, the retained route is the OPC UA server of the WinCC Runtime, read by a Python client in read-only mode.

Dataset	Scenario	Injection
D01	Five normal cycles with alternation	None
D02	Blocked flow on pump 1 until a reset	Low flow on pump 1
D03	Pump 1 draws more current than pump 2	High current on pump 1
D04	High current and low flow together	Both on pump 1
D05	Drain larger than the pump flow	Drain of 13 m^3/h
D06	Narrow level band, short restart delay	Level 48 to 52 %

Table 10.1. The six datasets. Ground-truth columns are kept apart from the measurements.

10.2 Reliability indicators and dashboard

Indicators are time-weighted sums over the intervals of the simulated clock, so a paused clock or a gap in the samples distorts nothing. Availability has two levels: with a backup pump a pump failure does not stop the service, and the gap between the two measures the value of the backup.

Indicator	Formula
Duty	$100\, t_{run} / t_{total}$
Start rate	$(n_{P1} + n_{P2}) / t_{hours}$
Station availability	$100\,(1 - t_{fault} / t_{started})$
Pump availability	$100\,(1 - t_{unavailable} / t_{total})$
MTBF, MTTR	$t_{run} / n_{failures}$, mean duration of the completed repairs
Mean flow while pumping	$\sum Q\,\Delta t \,/\, \sum \Delta t$ over the pumping intervals
Volume, energy, specific energy	$\sum Q\,\Delta t / 3600$, $\sum P\,\Delta t / 3600$, E / V

Table 10.2. Indicator definitions.

The computation has a unit test checked against a calculation done by hand (3 tests pass on the test suite).

Dataset	Station avail. (%)	Pump 1 avail. (%)	Starts/h	kWh/m^3
D01	100.0	100.0	0.417	0.491
D02	100.0	41.04	0.501	0.491
D03	100.0	100.0	0.835	0.565
D04	100.0	5.00	0.572	0.510
D05	100.0	100.0	0.201	0.491
D06	100.0	100.0	5.641	0.491

Table 10.3. Indicators of the six datasets (from `kpi_summary.md` and `kpi_summary.csv`).

The gap between station availability and pump 1 availability in D02 is 59 points: the station stayed at 100 % because pump 2 took over when pump 1 was faulted. That gap is the value of the backup pump. The same gap appears in D04 where pump 1 is unavailable almost the whole run.

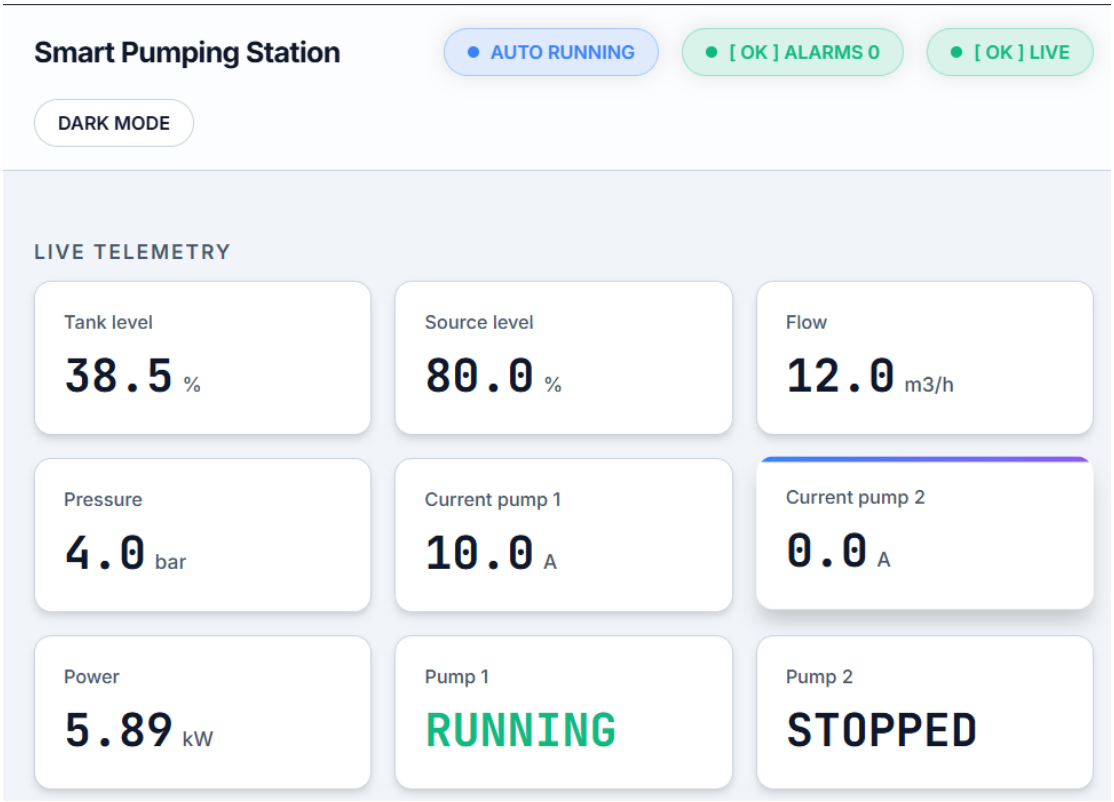


Figure 10.2. Local read-only dashboard in live mode.

The dashboard is a Python standard-library server on the local address. It only answers read requests, loads no external resource, and a pill shows STALE when no new sample has arrived for five seconds.

10.3 Explainable anomaly rules

Six rules read the measured signals only. Each flag gives the values and the threshold crossed. The injected-fault columns are used by the evaluation module only, and a unit test checks that the rules never name them. No machine learning is used: the roadmap makes it conditional on a documented limit of the rules.

Rule	Condition	Persistence
ANO_01	Pump running and flow below the minimum	5 s
ANO_02	Same pump running for the window, level rise under 0.5 %/min	5 simulated min
ANO_03	Motor current above 12 A with low flow	5 s
ANO_04	More than 3 starts in the trailing window	60 simulated min
ANO_05	Value out of range, or inconsistent state	range: none, states: 2 s
ANO_06	One pump draws over 15 % more current than the other	settled 2 s

Table 10.4. Anomaly rules.

Thresholds shared with the PLC keep the PLC value. The analysis delay of 5 s is half of the 10 s PLC validation delay, so the rule sees the symptom before the protection trips. The margin of every threshold is measured on the normal dataset, and a sensitivity table shows what changes when a

threshold is halved or doubled.

Dataset	ANO_01	ANO_02	ANO_03	ANO_04	ANO_05	ANO_06
D01	ok	ok	ok	ok	ok	ok
D02	ok	ok	ok	ok	ok	ok
D03	ok	ok	ok	ok	ok	TP
D04	ok	ok	ok	ok	ok	ok
D05	ok	TP	ok	ok	ok	ok
D06	ok	ok	ok	TP	ok	ok

Table 10.5. Rule outcome per dataset (TP: expected rule fired, ok: neither expected nor fired).

Two rules illustrate the result. ANO_06 fires on D03 with 63 flagged samples and 2 periods, with a mean detection delay of 2 samples. ANO_02 fires on D05 with 258 flagged samples and 1 period, with a mean delay of 6 samples (6 min). ANO_04 fires on D06 with 192 flagged samples. The documented blind spots are the reason why D02 and D04 show no firing: the injection flag forces the PLC to trip its low-flow protection but does not modify the recorded `Flow_m3h`, so the recorded data never shows "pump running AND flow < 2". In the field, where the flow sensor is the detector, the rules would fire.

Dataset	Rule	Fault events	Detected	Mean delay
D03	ANO_06	2	2	2 samples (2 min)
D04	ANO_06	1	0	n/a (blind spot)
D05	ANO_02	1	1	6 samples (6 min)

Table 10.6. Detection against the injected faults (from `anomaly_eval.csv`).

Check	Extreme on normal data	Threshold
ANO_01 lowest flow while pumping (m^3/h)	12.0	2.0
ANO_03 highest current while pumping (A)	10.0	12.0
ANO_02 lowest level rise (%/min)	1.47	0.5
ANO_04 most starts in the window	1	3
ANO_06 largest excess current (%)	0.0	15.0

Table 10.7. Threshold margins on the normal dataset.

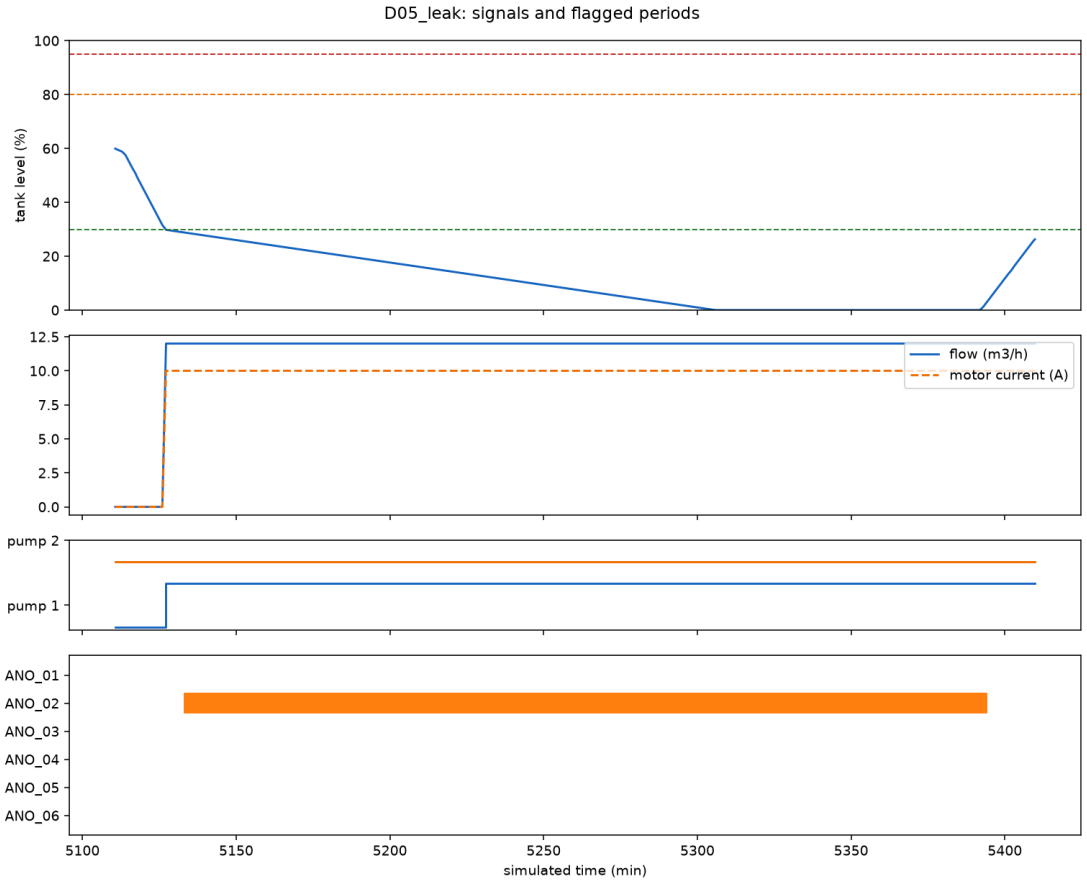


Figure 10.3. Signals and flagged periods for the leak dataset.

CHAPTER 11

Test plan and results

11.1 Method

Thirty-two tests were written before the PLC code and the plan grew to 69 tests with the derived requirements. The final campaign replays every manual PLC and HMI test on a frozen build, with only defect fixes allowed between the start of the campaign and the release tag. After a fix, the tests to rerun are taken from the traceability matrix. Three evidence levels apply: captures or videos for safety and end-to-end tests, observed values for the others, and automatic checks with their output saved.

Group	Tests	Passed	Failed	Pending	
PLC logic	31	31	0	0	
HMI	14	14	0	0	
Data	7	7	0	0	
Analytics	6	6	0	0	
Process and reproducibility	6	6	0	0	
Acceptance	5	5	0	0	
All	69	69	0	0	

Table 11.1. Results of the final campaign (from `campaign_summary.md`).

Requirements verified: 56 of 56.

11.2 Acceptance scenarios

The five use cases of the design phase are replayed end to end: HMI commands, PLC simulation and the OPC UA collector recording a dataset.

Test	Use case	Expected result	Result
T61	Normal start in Auto	Priority pump starts at the low level, feedback confirmed, level rises	Pass
T62	Fill and stop	Pump stops at the high level, priority passes to the other pump	Pass
T63	Pump failure	Pump locked out, backup takes over, ALM_006	Pass
T64	Source shortage	Pump stops, starts inhibited until recovery, ALM_002	Pass
T65	Low flow	After 10 s the pump is unavailable, backup used, ALM_004	Pass
T68	HMI E-stop single press	Pump stops at once, FAULT, ALM_001, no confirmation window	Pass
T69	HMI E-stop release	Cancel keeps the stop, confirmation clears it, reset gives STOPPED	Pass

Table 11.2. End-to-end acceptance tests.

11.3 Traceability and defects

Requirement	Tests	Subject
REQ-011	T08	Emergency stop latches the station in FAULT
REQ-014	T09, T12	Reset refused while the cause persists
REQ-012	T15	Critical tank level
REQ-045	T23	Acknowledgment does not clear a fault
REQ-054	T58	Rules never read the injected-fault columns
REQ-056	T68, T69	HMI emergency stop with confirmed release

Table 11.3. Examples from the traceability matrix.

ID	Description and resolution	Severity
BUG-001	Clock tag undefined: clock memory byte enabled in the CPU	Minor
BUG-002	Instance data blocks out of date after a block interface change: update order documented	Minor
BUG-003	Communication driver error of the HMI connection: HMI connection misconfigured after the CPU rename, recreated with the correct name	Minor
BUG-004	External S7 reader cannot reach the simulated S7-1200: route changed to OPC UA	Major
BUG-005	OPC UA port closed: runtime package, certificate authority and IIS set up, sequence documented	Major

Table 11.4. Defects found by execution.

Four more findings came from design review, before any test: five requirements missing from the first version (state machine), a no-pump alarm that needed two resets (cause based on latches), a pump command that survived a mode change (HMI interface), and a start window counted in real time while the process ran accelerated (data design). Each became a requirement or a fix with a test.

No defect was found during the final campaign on the frozen build.

11.4 Reproducibility

The cold start procedure of the runbook was run twice with identical initial values (T28, 15 variables checked in both runs), the project was restored from its release archive in a new folder with zero compilation errors (T66), and a clean clone of the repository passes the automated verification with all 20 checks in final mode (T67).

CHAPTER 12

Limitations and perspectives

Limitations

- Validation is entirely in simulation: S7-PLCSIM and WinCC Runtime, with no physical controller or sensor, and a process simulated inside the PLC.
- The indicators and the datasets come from the simulation, with one or two injected failures: they illustrate the calculation and are not field statistics.
- The emergency stop and the latched faults are simulated logic, not a certified safety chain. The HMI emergency stop request is a supplementary action, the reference emergency stop remains the hardware contact.
- No IEC 62443 compliance is claimed. The OPC UA endpoint accepts anonymous access on the laboratory PC only.
- Thresholds come from a noiseless simulation: their margins have to be measured again on field data.
- The rules have documented blind spots. The comparison between the two pumps cannot judge the first run of a dataset and does not say which pump deviates, and a level that does not rise does not distinguish a leak from a flow sensor error. The D02 and D04 datasets show a limitation of the injection itself, which bypasses the recorded flow.

Perspectives

- Secure the OPC UA endpoint with signed and encrypted messages and certificate trust management, then assess the security by zones and conduits.
- Port the project to an S7-1500 to use S7-PLCSIM Advanced and a native OPC UA server.
- Detect leaks with a mass-balance residual, the level change compared with the integrated flow.
- Evaluate a statistical or learned detector against these rules on field data.
- Move the HMI to a physical panel with user profiles, and add a hardware-in-the-loop test bench.

General conclusion

The project delivers a complete chain on a simulated pumping station: requirements and tests written before the code, a modular PLC program with latched faults and a conditional reset, a two-pump station with alternation and backup, an HMI with a native alarm system and an emergency stop whose release is confirmed, a read-only data path, reliability indicators and six explainable rules, all verified by a campaign of 69 tests on a frozen build with no failure.

Five lessons stand out:

1. Writing the tests first forces the requirements to be clear: the design of the state machine alone revealed five missing requirements.
2. A feasibility test before building avoids wasted work: the planned data route was impossible and the test showed it in minutes.
3. Two clocks need two rules: delays on the real clock, dynamics on the simulated one, otherwise a false alarm hides in the accelerated runs.
4. Traceability is a decision tool: after a fix, the matrix says what to rerun, and a shared block triggers the full level 1 regression.
5. A threshold without a measured margin is a guess: every rule threshold was checked against the normal dataset.

The emergency stop addition at the end of the project shows the same process at work: a design question, a decision justified by safety reasoning (no confirmation on press, confirmed release), a new requirement, two new tests, an impact analysis and a full level 1 regression before the tag. The change is delivered with the project, not as an afterthought.

References

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[5] ISO 13849-1:2023. *Safety of machinery, Safety-related parts of control systems, Part 1: General principles for design*. International Organization for Standardization, 2023.

[6] ISA-18.2:2016. *Management of Alarm Systems for the Process Industries*. International Society of Automation, 2016.

[7] Siemens AG. *SIMATIC S7-1200 System Manual*. Siemens Industry Online Support.

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[9] K.-H. John and M. Tiegelkamp. *IEC 61131-3: Programming Industrial Automation Systems*. 2nd ed. Springer, 2010.

APPENDIX A

Repository structure

```
smart-pumping-station/
README.md, README.fr.md, CHANGELOG.md, CONTRIBUTING.md, LICENSE
01_requirements/   scope, assumptions, requirements, use cases, traceability matrix
02_design/         P&ID, I/O list, alarm matrix, state machine, diagrams/
03_plc/            src/ (SCL sources), exports/, TIA Portal notes
04_hmi/            screens/, operator guide, design standard, tag and alarm lists
05_iiot_data/      datasets/, python/, dashboard/, figures/, notebooks/, results
06_tests/          test plan, results, campaign, defect log, evidence/
07_media/          demonstration script and video notes
08_report/         report_final.tex, report_final.pdf
docs/              process, parameters, runbook, decision log, glossary
tools/             figure generation, campaign, verification, impact analysis
captures/          screenshots by iteration
```

APPENDIX B

Main parameters

Parameter	Default	Role
`Level_Low`	30 %	Start threshold
`Level_High`	80 %	Stop threshold
`Level_CritHigh`	95 %	Critical high level
`Source_Min`, `Source_Hyst_pct`	15 %, 5 %	Source low threshold and recovery hysteresis
`Flow_Min`	2.0 m^3/h	Low flow threshold
`Flow_Validation`	10 s	Delay before judging the flow
`Restart_Delay`	30 s	Anti-restart delay
`Run_Feedback_Timeout`	5 s	Feedback expected after a command
`Init_Delay`	2 s	Duration of the initialization
`Maint_Test_Max`	60 s	Longest pump test in maintenance
`FreqStarts_Max`	3 per 60 min	Frequent start threshold
`Current_Nominal`, `Current_High`	10 A, 12 A	Nominal current and warning threshold
`Current_High_Delay`	5 s	Ignore the start-up inrush
`Maintenance_Interval_h`	500 h	Service interval
`Raw_Low_Limit`	-500	Wire break limit, about 3.7 mA
`LevelRise_Window`, `LevelRise_Min`	60 s, 0.5 %	Level rise warning
`Tank_Volume`	10 m^3	Level model
`Pump_Flow`, `Drain_Flow`	12, 3 m^3/h	Level model
`Sim_Step`	0.1 s	Euler step, equal to the interrupt period
`Time_Accel`	1 or 60	Acceleration of the process model
`Supply_Voltage`, `Power_Factor`	400 V, 0.85	Simulated electrical power

APPENDIX C

Alarm list

ID	Class	Condition	Reaction
ALM_001	Critical	Emergency stop contact open, or HMI request engaged	All pumps off, FAULT latched
ALM_002	High	Source low signal, or source level under 15 % (recovers at 20 %)	Pumps stop, starts inhibited
ALM_003	Critical	Tank level above 95 %	All pumps off, FAULT latched
ALM_004, 005	High	Pump 1 or 2 running, flow under 2 m^3/h for 10 s	That pump unavailable, latched
ALM_006, 007	Medium	Pump protection open, or no run feedback within 5 s	That pump unavailable, latched
ALM_008 (tank)	Critical	Tank level signal invalid	FAULT latched
ALM_008 (other)	Warning	Flow, pressure, source level or current signal invalid	Information
ALM_009	Critical	Demand and no pump available	FAULT latched
ALM_010	Warning	More than 3 starts in 60 simulated minutes	Information
WRN_002 (P1, P2)	Warning	Current above 12 A for 5 s	Information, pump keeps running
WRN_003 (P1, P2)	Warning	Run hours since service above 500 h	Information
WRN_004	Warning	Pump running, level rise under 0.5 %/min over 60 s	Information